

Outline

The goal of this research project is to construct a model that predicts daily stock trading volume surprises, using the 60-day half-life EWM as a benchmark. I divided the research into 3 phases, each utilizing different subsets of the dataset.

Phase 1: feature construction (2000-01-01 to 2010-01-01)

Using only the first 10 years of data:

1. I perform descriptive studies of the target variable and construct a set of baseline features.
2. Using LGBMRegressor with a fixed seed and gamma objective, I run a walk-forward train/predict at a yearly frequency on the 10 years of data. The predictions are then scored based on the R^2 in a global pooled regression in the 10 years of data.
3. I then incrementally tested new ideas through an LLM agentic procedure:
 - a. A skill is created to provide context on the problem
 - b. The LLM refers to a file for past tested ideas, then suggest new ideas and implement them in a Python module in a specified format
 - c. A harness is used to generate the feature using the Python module, and evaluate its incremental effect on the model's performance using the same walk-forward procedure
 - d. The results are then logged into the file for the next iteration.

The details are in `.claude/skills/volume-researcher/SKILL.md`

Phase 2: model construction (2000-01-01 to 2015-01-01)

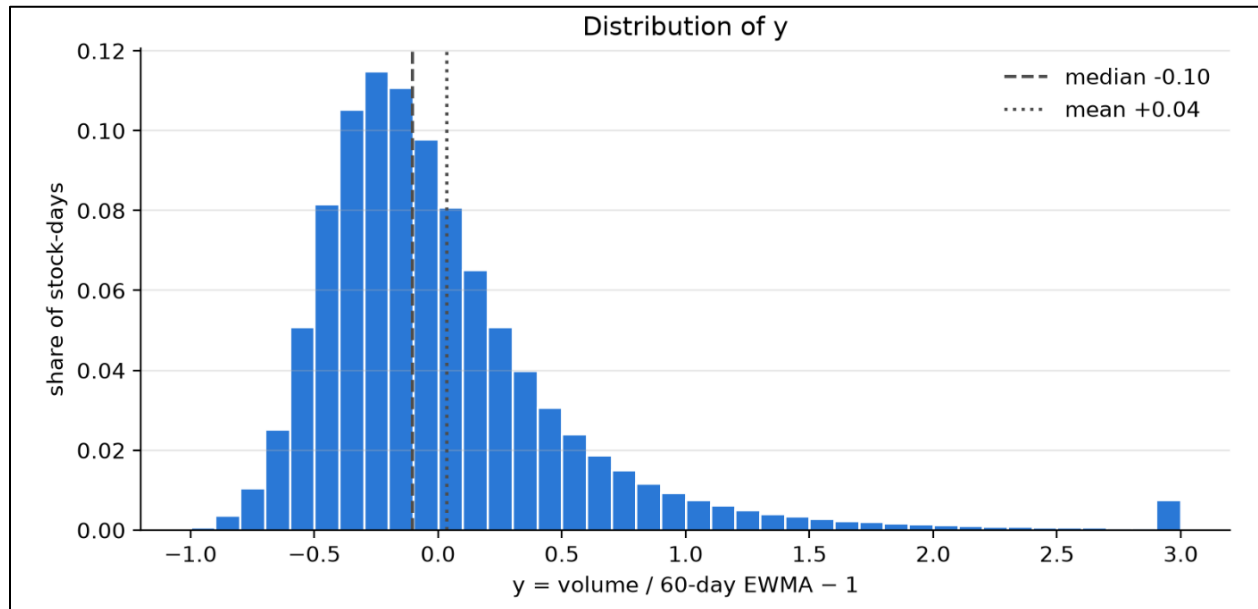
Having decided on the set of features, I used the entire in-sample period to run model selection. Optuna is used to perform the search, and the cross-validation score is computed using the walk-forward predictions for the entire 15 years.

Phase 3: out-of-sample (2015-01-02 to 2020-01-01)

Having decided on the features *and the model*, the entire walk-forward procedure runs on the entire 20 years, and the predictions in the last 5 years are scored.

Q1: Descriptive Study (uses only first 10 years of data)

Pooled target

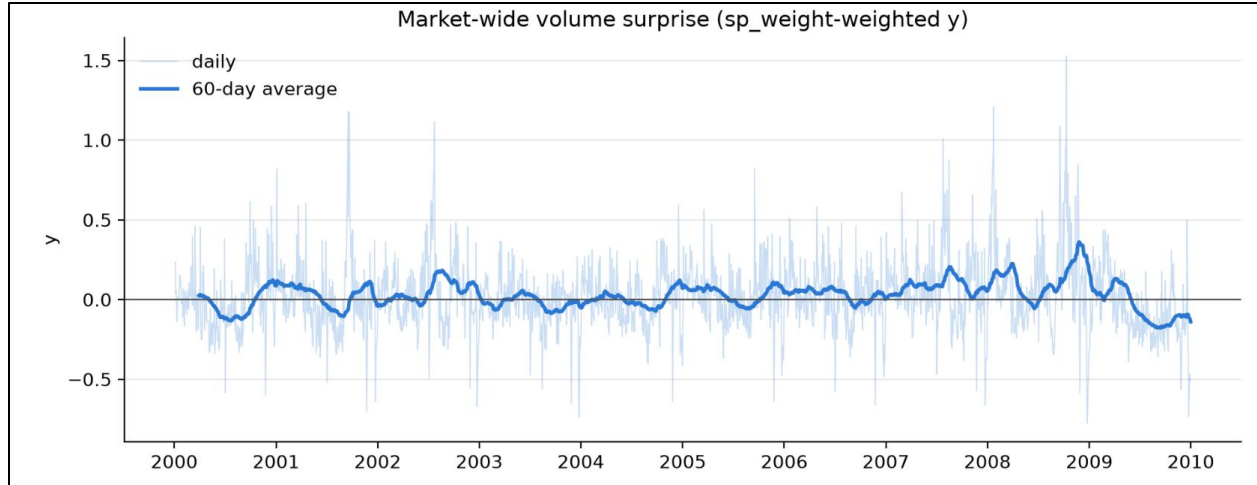


The above chart shows the distribution of y (clipped at 3 for visualization). A few observations can be made:

- **Typical days are quiet.** mode < median < 0 < mean.
- **A small number of days have abnormally large volume.** The skew is 5.4, and about 5% of the days have $y > 1$ i.e. double a stock's past trading volume. It is likely that these are earnings and other news days. The fraction of variance attributed to these 5% of events is about 68% on a weighted basis, as calculated in this way:

$$\text{share}_{y>1} = \frac{\sum_{i: y_i > 1} w_i (y_i - \bar{y}_w)^2}{\sum_i w_i (y_i - \bar{y}_w)^2}, \quad \bar{y}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

Market-wide

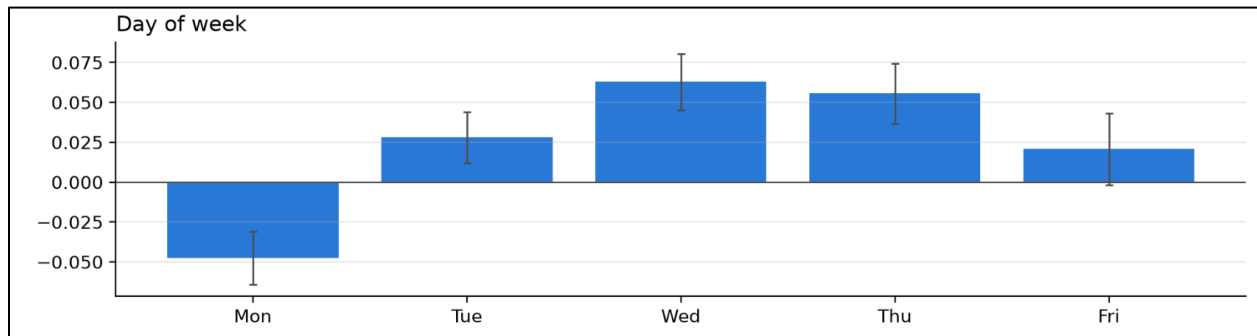


The above chart shows the sp_weight-weighted y . The purpose is to understand, on aggregate, the distribution of y .

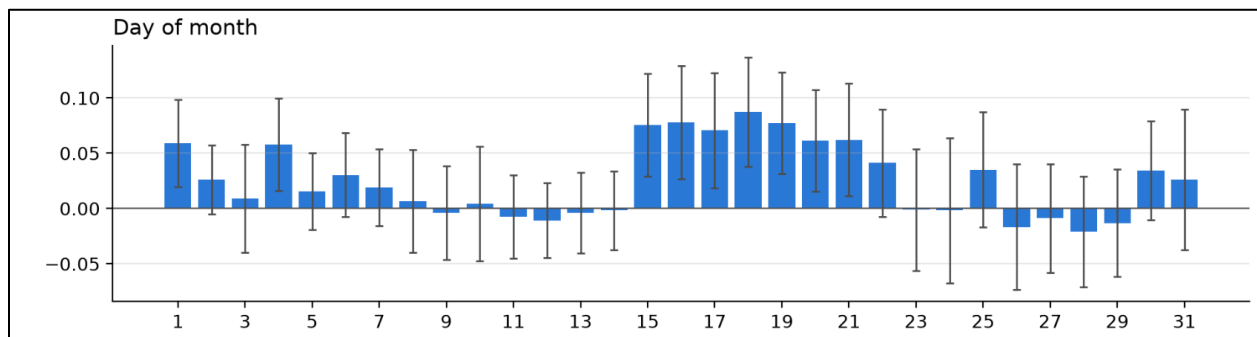
- The period in the second half of 2008 (financial crisis) shows a large positive y , followed by large negative y . Intuitively this is caused by estimating volume with a long half-life; when a volume shock hits, the trailing long half-life will always underestimate, and after the volume shock ends, the trailing long half-life always overestimates. This suggests adding shorter-term trailing volume surprise as factors.

Calendar Effect

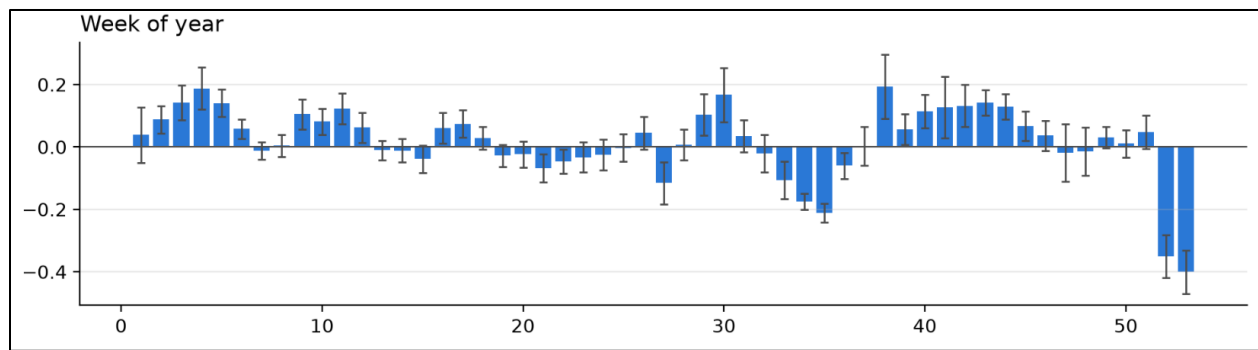
The sp_weight-weighted mean for each day is first computed (as in the previous section), then the daily time-series is grouped by days, weeks, months etc. to understand the various calendar effects. The standard error is computed using the daily time-series, not pooled across all rows, because on each day there is a common market component for the volume and so rows on the same day are not independent.



- Mondays generally have lower volume surprises. Fridays are also slightly lower, while the middle of the week is highest. This is probably because most earnings events happen near the middle of the week, and Fridays usually have the least earnings. However, options expiration and index rebalance usually happen on Fridays, mitigating its lack of trading volume somewhat.



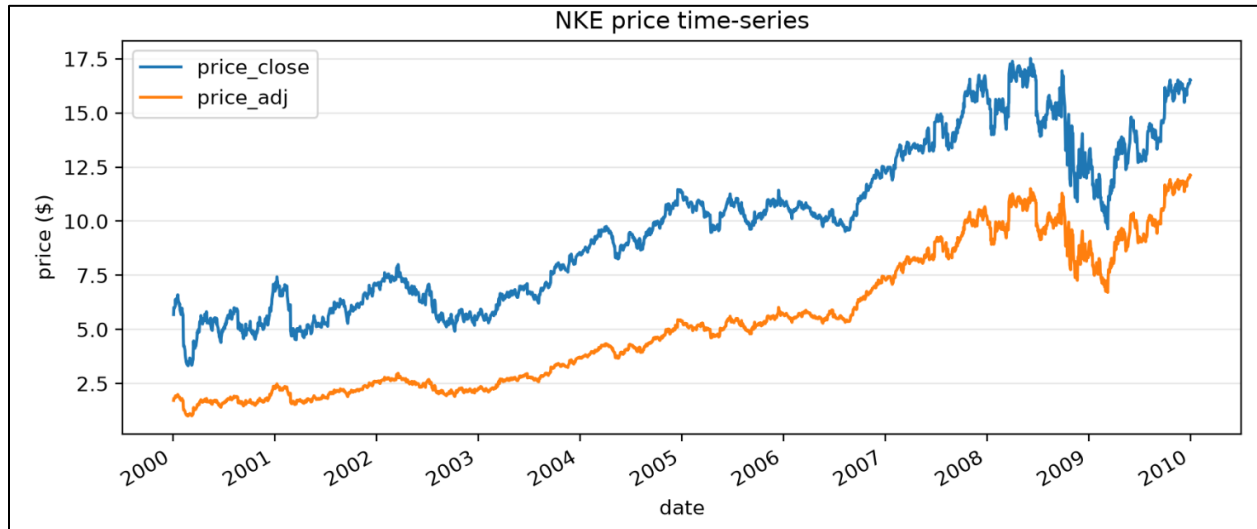
- Most of the volume surprise is concentrated around the third week, probably because of index rebalance/options expiration.



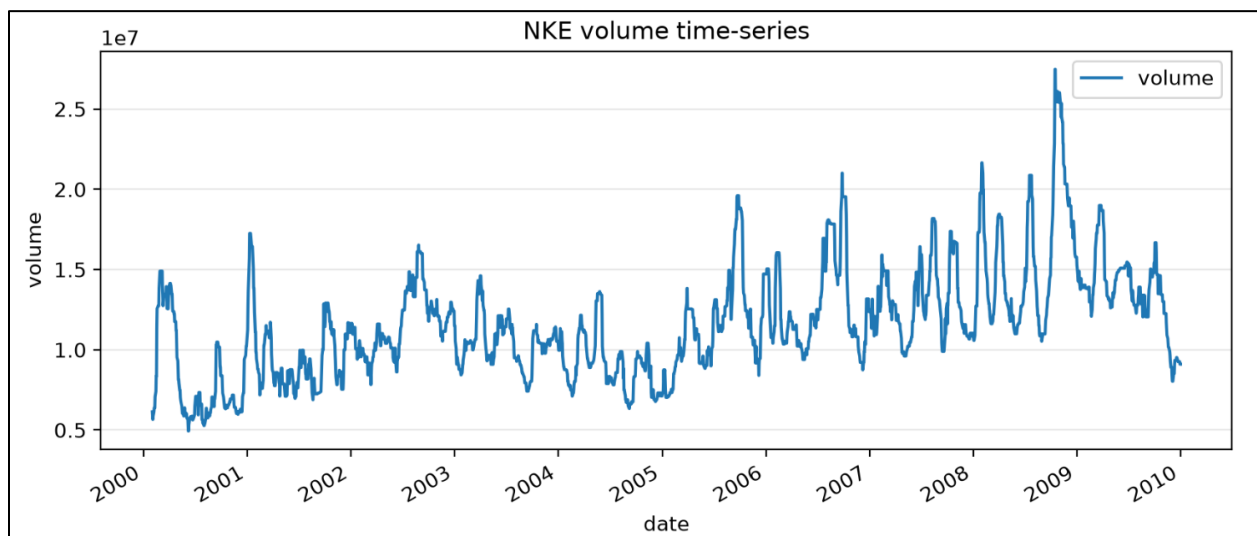
- The final weeks of the year show much less volume surprise than usual, possibly because of holidays and early closing.

Corporate Action

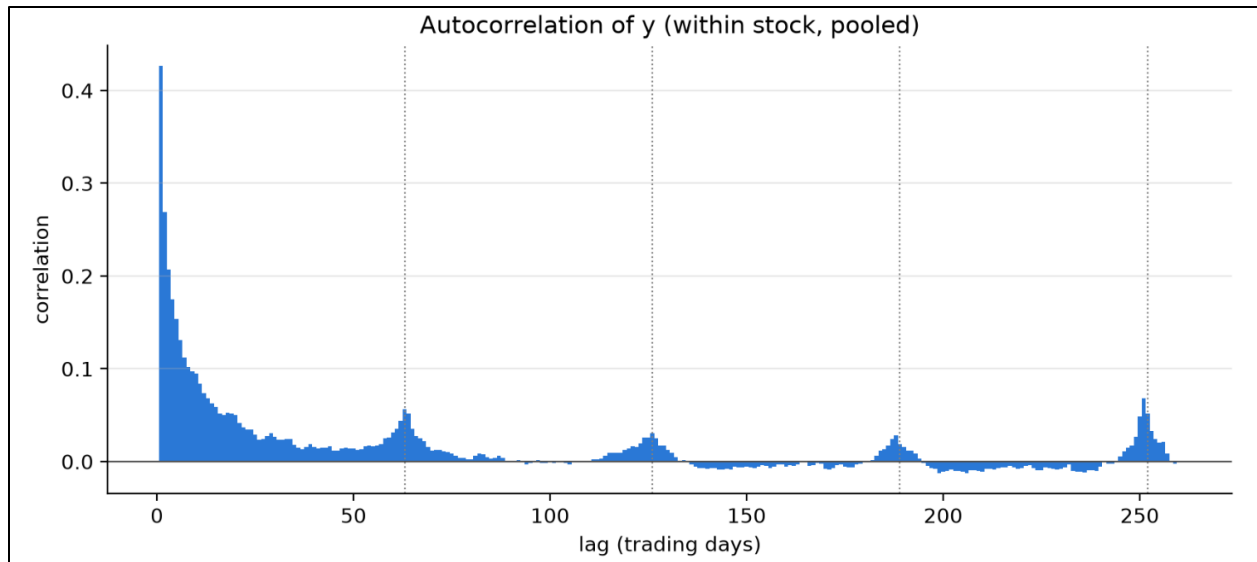
If the data provided has not been adjusted for corporate action such as stock-splits, then on such days it would make very inaccurate volume predictions. However, after looking at the data I can see that `ret_raw` is computed from percentage changes in `price_adj` (a simple computation confirms this), and both `price_close` and `price_adj` are at least split-adjusted. Looking at NKE as an example, it had stock split(s) in the 2000 to 2010 period, but the price time-series show no jump:



Similarly, the volume time-series shows no persistent increase at any time, as would be expected if a stock split were to happen.



Autocorrelation



At each lag from 1 to 260, compute the pooled autocorrelation of the target y . For short lags it is clear the volume persists. More interestingly, there are 4 peaks happening around the 63, 126, 189, 252 lags (the vertical dotted lines). These are most likely the past 4 quarters of earnings. This clearly suggests that being able to predict earnings days should be a feature.

Q2. Model

The model is evaluated using a global univariate regression and a good model should have a large R^2 with a positive coefficient and high t-stat when tested out-of-sample. I would assume that the regression is done with an intercept, because the R^2 would exactly be the squared weighted correlation of my prediction and Y, and maximizing R^2 is equivalent to maximizing correlation. Furthermore, **with a fixed dataset**, there is a monotonic relationship between R^2 and t-statistics. So, maximizing R^2 is equivalent to maximizing t-statistics.

To maximize correlation with Y would be to find a model that predicts the conditional expectation of Y given the data $E[Y|X]$. A mean-squared-loss is a loss function that satisfies this condition, but given that the target Y is right skewed, the effect of factors on the target is probably multiplicative and $Y+1 \geq 0$, another possible loss is gamma loss.

To model non-linearity and interaction between factors, I will use LightGBM's LGBMRegressor with a gamma objective as the model. It requires the target to be strictly positive, so in my training dataset I will ignore any zero-volume rows, and I will use $Y+1$ as the target.

I divide the dataset into 3 periods

1. feature construction: 2000-01-01 to 2010-01-01
2. model construction: 2000-01-01 to 2015-01-01
3. out-of-sample: 2015-01-02 to 2020-01-01

The feature construction period is used to run walk-forward fit/predict (at a yearly frequency). The model construction period is used for hyper-parameter tuning of the model, and the last 5 years is considered out-of-sample and evaluated at the very end.

Feature construction

An initial set of reasonable and intuitive features are constructed and with a walk-forward fit/predict procedure, the performance is evaluated. New features are then incrementally added to the model and tested if any improvement is observed. The final set of features are:

- **Volume persistence:** EWMA of past volume relative to the 60-day naive forecast (halfives 1, 5, 10, 21 days), lagged y (1–5 days), and rolling max of y (5, 10, 20 days).
- **Volatility and returns:** EWMA return volatility (halfives 1, 5, 10, 21 days) and lagged returns (1–5 days).

- **Earnings schedule:** probability that today is an earnings day, projected from the stock's past quarterly volume spikes 13, 26, 39 and 52 weeks earlier (and adjusting for Q1 financial year-end offset), with a separate factor using only spikes confirmed by an overnight price gap.
- **Price path:** 20- and 60-day cumulative returns, and position within the 52-week high–low range.
- **Overnight vs intraday:** the previous day's return split into its overnight gap and intraday move, each scaled by volatility.
- **Calendar:** day of week, month, week of month, month/quarter start and end, MSCI rebalance, options expiry, quad witching, and post-holiday flags.

Model Selection

With the set of features fixed, the first 15 years is used for hyper-parameter tuning of the model. The first 10 years have been tested extensively during feature construction, and therefore some degree of overfitting on the features and model already occurred. Selecting the hyper-parameters by adding the new 5 years of data can help mitigate this.

Optuna is used to perform the hyper-parameter search. Each cross-validation score is the output R^2 of a walk-forward run from 2000 to 2014. 100 trials are conducted.

Out-of-sample

The selected model and features are then run in a walk-forward procedure throughout the entire 20 years. The final 5 years of out-of-sample data is used for evaluating the model's predictions. For comparison, the default model also ran out-of-sample to see if the model selection procedure added any value.

Results

model	period	R^2	t-stat
default	in-sample	0.3117	807.3
default	out-of-sample	0.3554	584.1
tuned	in-sample	0.3163	815.9
tuned	out-of-sample	0.3630	593.8

- Both models generalized well out-of-sample, with the tuned model out-performing the default model. In fact, out-of-sample R^2 is even higher than in-sample, possibly because the last 5 years of data do not include volatile events like the Financial Crisis and its aftermath.
- The t-stat for the out-of-sample period is lower than in-sample only because the sample size is smaller out-of-sample.

Q3: Other ideas

Idea 1: Event Data

Data

Earnings dates are usually known many days in advance and both historical point-in-time and real-time data are widely available from external vendors/web-scraping.

Macro releases such as FOMC/CPI/jobs report dates are made public months ahead of time. Available from both web-scraping and external vendors.

Index Rebalance days such as Russell Reconstitution, MSCI rebalance, S&P rebalance are some of the highest trading volume days of the year. Rebalance events are usually announced by index providers days or weeks in advance. Both the announcement date (especially for S&P because additions/deletions are discretionary) and rebalance date usually see heightened volume.

Ad-hoc Events such as drug-trial result release, investor's conference etc. These events are not known from a single data source but can be inferred from implied volatility data by looking at the vol term structure.

Half-days typically around holidays like July 4th and Thanksgiving. Since market stops trading at 1pm, the trading volume is also expected to be lower. The Python package *exchange_calendars* provides such information, but I have decided to only use the provided data and calendar rules in this research.

Intuition

Stock trading volume increases when there is new information arriving from earnings or macro releases. Index rebalance forces buying/selling of stocks. A 60-day EWM cannot forecast these events in advance.

Features

- Indicator variable for each event
- Shrunk target encoding

$$x_{s,k} = \lambda_s \bar{y}_{s,k} + (1 - \lambda_s) \bar{y}_{\text{all},k}, \quad \lambda_s = \frac{n_s}{n_s + m}$$

Where $\bar{y}_{s,k}$ is stock s 's historical mean target on days to earnings equal to k , and $\bar{y}_{\text{all},k}$ is the historical mean target for all stocks on days to earnings equal to k . λ_s is

the shrinkage parameter, n_s is the number of past events for stock s and m is the tuning parameter. The intuition is that each stock reacts differently to earnings and so the stock's own historical average matters. However, each stock's history is limited and the data is noisy, therefore a shrinkage towards market-wide mean is necessary. The use of days to earnings is potentially informative because volume increase is not only present on the earnings day itself; the trading volume is high also before and after the earnings announcement. The tuning parameter can be tuned on a validation set. A different λ_s is used for different stocks because for stocks with shorter history, the data is noisier and so the shrinkage towards the market-wide mean should be stronger.

Other calendar events can be encoded in a similar way. For example, FOMC and CPI release might affect different stocks differently depending on their interest rate sensitivity. Therefore, a shrunk target encoding is also suitable.

All averages should only use data in a past window to compute to avoid look-ahead bias.

- Index-rebalance flow: For index rebalance, the fraction of dollar flow over a stock's average-daily-traded volume can be added as an additional feature. The dollar flow can be estimated by the index AUM multiplied by the stock's weight change in the index.

Idea 2: Overnight news and pre-open return and volume

Data

Various vendors provide real-time news feeds coming from news wires, SEC EDGAR filings, web scraping.

Intuition

Real-time news data provide new information on stocks and drives stock trading volume. Certain categories of news events have especially strong effect on stock trading, such as analyst revisions, earnings preliminary announcements, FDA drug trial results, lawsuits, M&A rumor/news, activist investor activities, strategic investment partnerships etc.

Features

An NLP model like FinBERT can be used to generate features from the text data, and the features are then fed into a downstream model such as LightGBM. To use FinBERT, the

lower layer weights can be frozen and the top classifier head can be replaced with a regression head that predicts y . Then with a low learning rate the model can be fine-tuned on the financial news data.

Directly downloading and using pre-trained models would incur some look-ahead bias because most such models were trained using recent data. However, it is likely that such a bias is small for the purpose of volume predictions because the model was initially trained on sentiment classification.

Since predictions are made before the market opens, one can also use pre-open stock volume and overnight return information. This would probably be one of the strongest predictors.